Paper A v1.0-candidate — The Physical (SU(3)) Yang–Mills Mass Gap via D_IV^5 Spectral Geometry. Companion to Paper B. v1.0-candidate = v0.9 corrected to DIMENSION-ROOTED (K1713). The gap value is CLOSED and referee-safe: reading the field correctly — Yang–Mills is the gauge field, a 1-FORM — fixes the operator as the FORM (Hodge) Laplacian; transverse 1-forms on the boundary sphere S^{d_S} start at 2(d_S−1) = 6, set by the SPHERE DIMENSION d_S = n_C−1 = 4 (NOT the color number — U(1)/SU(3)/SU(5) all give 6; the gauge group sets the DEGENERACY, not the eigenvalue). Form parity (k+p+m even) keeps the pure-spatial (1,0) mode; gauge invariance eats A_θ → physical gluon (1,0), m=0 → mass^2 = 2(d_S−1) = 6, R-INDEPENDENT, dimension-rooted, DERIVED on the compact boundary. LABEL DISCIPLINE (Guard 3): 2(d_S−1), NEVER ‘colors’ and NEVER ‘= C_2’ (scalar Casimir n_C+1, the morning’s identity K1707). The Δ→mass^2 shift is CLOSED (no conformal shift; the transverse eigenvalue is the physical mass^2). THREE NAMED RESIDUALS: (1) the flat-$\mathbb{R}^4$ construction (Clay); (2) the literally-YM identification (interacting, not generalized-free); (3) the constant-holonomy zero mode (finite-volume modulus, declared). Scalar readings (4, §570 (1,1)) = wrong sector, superseded. Polyakov mode at 10 above the gap (deconfinement tie). Glueball 1720 separate-and-valid (on C_2^int), out of the room. The value is closed; the package is the deliverable. NOT 'YM proved', NOT 'colors', NOT '= C_2'. Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_IV^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v1.0-candidate — the gap value is CLOSED and dimension-rooted (K1713): the gauge field is a 1-form, so the operator is the form (Hodge) Laplacian, whose lowest transverse eigenvalue is 2(d_S−1) = 6, set by the boundary-sphere dimension d_S=4 (not colors — U(1)/SU(3)/SU(5) all give 6); the pure-spatial (1,0) gluon gives mass^2 = 2(d_S−1) = 6, R-independent — derived. Labeled 2(d_S−1), never "colors," never "= C_2". Δ→mass^2 shift closed. Three named residuals: the flat-$\mathbb{R}^4$ construction, the literally-YM identification, and the constant-holonomy zero mode. Glueball separate-and-valid, out of the room. The value is closed; the package is the deliverable.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is a spectral gap on the compact boundary of D_IV^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. mass^2 = 2(d_S−1) = 6 on our boundary is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don't assert a shared eigenvalue). *(This is residual (1), the flat-$\mathbb{R}^4$ construction.)*
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object); we do not lend the proton's four-decimal precision to the Yang–Mills gap (K1705). And the glueball is *out of the room* for the gap-*value* pin (K1708/K1709): 1720 = (11/6)·938 is built on C_2^int (the labeled integer primary), not on λ_1 (the boundary Laplacian gap), and its 11/6 carries C_2 = 6 — so it is a real, separate forward prediction but not independent evidence for the gap value. The value is pinned from the geometry alone (the GATE-KOIDE discipline that keeps Q = 2/3 out of the Koide pin).
- Guard 3 (the symbol "C_2" itself — K1707). Three distinct objects all read "C_2" and all equal 6 at the physical point n = n_C = 5, so writing "the gap = C_2 = 6" hides *which one* is meant:
  - C_2^int := n_C + 1 = 6 — the *labeled integer primary* (one of the five integers). A definition.
  - C_2^flow(n) = 2n − 4 — the *commitment-flow Casimir* (a function of n; = 6 at n=5 only by the coincidence-at-n_C=5).
  - C_2^Berg = n + 1 — the *Bergman-representation Casimir* (a function; = 6 at n=5, and equal to C_2^int *identically*, since C_2^int is *defined* as n+1).

  The consequence (Cal, K1707): "λ_1 = C_2 = 6" as previously written is C_2^int = C_2^Berg — the same definition on both sides, an identity that *cannot fail* (P^2=P), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator's* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).
- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography → Osterwalder–Schrader) places on the flat 4D boundary literally interacting SU(3) Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

The result, and the residuals — in one sentence. On the compact boundary the Yang–Mills mass gap is a derived value — mass^2 = 2(d_S − 1) = 6, dimension-rooted (d_S = n_C − 1 = 4, the boundary sphere's dimension) and R-independent (the gauge field is a 1-form; transverse 1-forms on S^{d_S} start at 2(d_S−1); gauge invariance eats the time-component, leaving the pure-spatial (1,0) gluon) — with three named residuals: (1) the flat-$\mathbb{R}^4$ construction (that this compact-boundary gap is the Clay $\mathbb{R}^4$ Yang–Mills gap), (2) the literally-YM identification (the reconstructed edge is interacting SU(3) YM, not generalized-free), and (3) the constant-holonomy zero mode (a finite-volume modulus, declared). The $\Delta \to \text{mass}^2$ shift is now closed (K1713 — no conformal shift; the transverse eigenvalue is the physical mass^2).

So the paper's claim is explanatory, and now with a closed value: BST *derives* the Yang–Mills mass-gap value on its compact boundary (2(d_S−1) = 6), with the flat-space construction the named residual. The gap exists because the domain is bounded, its value is fixed by the geometry (no dimensionless knob), and the number is dimension-rooted — set by the boundary-sphere dimension, not "colors" (the gauge group sets only the degeneracy; U(1)/SU(3)/SU(5) all give 6) and not the scalar Casimir C_2 (Guard 3: same number, different object; "= C_2" was the morning's identity). Nothing here is "we solved Yang–Mills": residuals (1)–(2) are the large flat-$\mathbb{R}^4$ open, carried as such.

Abstract

We study the Yang–Mills mass gap for the physical gauge group SU(3) — color — derived (not assumed) from the substrate D_IV^5 in the companion Paper B. The mass-gap operator is a Laplacian on the physical Shilov boundary ∂_S $D\_IV^5 = (S^4 \times S^1)/\mathbb{Z}_2$; its gap is nonzero because the domain is bounded, with no free dimensionless parameter beyond the ruler R. The value is derived (K1713): $\text{mass}^2 = 2(d\_S - 1) = 6$, dimension-rooted and R-independent. The decisive step is reading the field correctly — Yang–Mills is the gauge field, a 1-form, so the operator is the form (Hodge) Laplacian, whose lowest eigenvalue on transverse 1-forms of the boundary sphere is $2(d\_S - 1) = 6$, set by the sphere dimension $d\_S = n\_C - 1 = 4$. It is dimension-rooted, not dimension-rooted: U(1)/SU(3)/SU(5) all give the same 6 — the gauge group sets the *degeneracy*, not the eigenvalue. On $(S^4 \times S^1)/\mathbb{Z}_2$ the form parity (k+p+m even) keeps the pure-spatial mode (1,0), and gauge invariance eats the time-component (A_θ pure gauge for m≠0), so the physical transverse gluon is (1,0), m=0 — R-independent. This 6 must never be called "colors" and never relabeled "= C_2" (the scalar Casimir n_C+1, a different object; the identity trap corrected earlier this session, K1707). The earlier scalar readings (bare-S^4 "4", the $\mathbb{Z}_2$-projection, the §570 (1,1) candidate) were the wrong sector, superseded by the form reading. The $\Delta \to \text{mass}^2$ shift is closed (K1713 — no conformal shift). Three named residuals remain: (1) the flat-$\mathbb{R}^4$ construction (the Clay map), (2) the literally-YM identification (interacting, not generalized-free), and (3) the constant-holonomy zero mode (a finite-volume modulus, declared). *(The glueball 1720 MeV = (11/6)·938 is a separate valid prediction built on C_2^int — kept, but out of the room as gap-evidence, K1708.)* The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_IV^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain's 4D conformal boundary (SO(4,2) ⊂ SO(5,2)), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis

on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* SU(3) YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^PC channels follow a linear conformal-energy ladder off one verified number (the genus of D_IV^5), with one blind leg, 2^{++} = g/n_C, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^PC tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat-$\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not "all G"

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: G = SU(3), color. The companion Paper B proves D_IV^5 is the unique substrate and that SU(3) is derived, not chosen. So "all G" is not BST's claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: value DERIVED on the compact boundary — mass^2 = 2(d_S − 1) = 6, dimension-rooted, R-independent (K1713); three named residuals

What is SOLID (existence). The mass-gap operator is a Laplacian on the Shilov boundary of D_IV^5 — a positive operator whose spectrum *is* the arrow of time. The domain is bounded, so the spectrum is discrete and bounded below; the gap is nonzero *because the domain is bounded* (decompactify, R → ∞, and the gap → 0), with no free dimensionless parameter beyond the ruler R (a Planck-unit — the same single input GR takes as G and QM as ℏ). *Existence is not in question and never was.*

The value closes once you ask what the field is (K1711). The morning's confusion came from treating the gap as a *scalar* Laplacian eigenvalue. But Yang–Mills is the gauge field = the adjoint, a 1-form — so the operator is the form (Hodge) Laplacian, not the scalar one. Three facts then close the value:

1. Transverse 1-forms on the boundary sphere start at 2(d_S − 1) = 6 — not the scalar's 4. The lowest eigenvalue of the Hodge Laplacian on co-closed (transverse) 1-forms of S^{d_S} is 2(d_S − 1), set by the dimension of the boundary sphere d_S = n_C − 1 = 4. It is dimension-rooted, not dimension-rooted: a U(1), SU(3), or SU(5) gauge field all give the *same* eigenvalue 6 — the gauge group sets the degeneracy (the dim of the adjoint), not the eigenvalue. And it passes the dimension-generic test: 2(d_S−1) = 6 requires d_S = 4 specifically

— *our* boundary dimension, not a formula that gives 6 at every dimension.

2. The form-$\mathbb{Z}_2$ keeps the pure-spatial mode. On $(S^4{\times}S^1)/\mathbb{Z}_2$ the identification is $k + p + m$ even (spatial level k, form degree p, time mode m). For the gauge 1-form ($p = 1$) the mode $(1, 0)$ has $k{+}p{+}m = 2$, even — kept. The form degree supplies the parity that the scalar lacked; nothing is projected out.
3. Gauge invariance eats the time-component. The A_θ (time-circle) component is pure gauge for $m \neq 0$, so it carries no physical mode; the physical transverse gluon is $(1, 0)$, $m = 0$, pure-spatial. Hence the gap is R-independent — no time-circle radius enters. *(The remaining A_θ constant mode, m = 0, is the constant-holonomy zero mode — a finite-volume modulus, declared as residual 3 below, not part of the gap.)*

$\Rightarrow$ mass^2 = 2(d_S − 1) = 6, R-independent, dimension-rooted — a derived value on the compact boundary.

> Label discipline (Guard 3, load-bearing — K1711/K1713). This 6 is 2(d_S − 1), set by the boundary-sphere dimension d_S = n_C − 1 = 4. It must never be called "colors" (the gauge group sets only the degeneracy, not the eigenvalue — U(1)/SU(3)/SU(5) all give 6) and never be relabeled "= C_2" (C_2 = n_C+1 is the *scalar* Casimir, a different object that merely equals 6 at n_C = 5; equating them was the morning's identity trap, K1707). Write 2(d_S−1); the gap is dimension-rooted. *(The coincidence 2(d_S−1) = 2(n_C−2) — which read as "2N_c" in the v0.9 draft — holds because d_S = n_C−1; but the eigenvalue is rooted in the sphere dimension, not the color number.)*

A check above the gap. The Wilson-line / Polyakov (holonomy) mode sits at 10, above the gap — consistent with the confined phase and tying to the deconfinement scale (Section 7 / the T_c work). The gap is the lowest, at 2(d_S−1) = 6.

How the earlier readings resolve (audit trail). The scalar "4" (bare S^4), the $\mathbb{Z}_2$-projection of it, and the §570 spatial-mode candidate (1,1) = $4 + 1/R^2$ were all the scalar sector — the wrong sector for a gauge field. Reading the field correctly (a form) supersedes them: the form (1,0) mode is pure-spatial and R-independent, so the R-dependence and the "which $\mathbb{Z}_2$-even mode" question both evaporate. The old "6" (conformal reading = C_2 = n_C+1) and "7" (genus) were never the gap.

reading	value	status
scalar S^4 Laplacian, k(k+3) first level	4	wrong sector (scalar); $\mathbb{Z}_2$-projected-out anyway
scalar §570 spatial-carrying mode (1,1)	$4 + 1/R^2$	wrong sector; superseded by the form reading
conformal reading / Bergman-rep Casimir, n+1	6	= C_2 — the scalar identity (K1707), not the gap
genus, g = n_C+2	7	a different invariant

reading	value	status
gauge-form (1,0), Hodge Laplacian	2(d_S−1) = 6	the gap — dimension-rooted (d_S=4), R-independent, DERIVED (K1713)

The glueball stays separate and out of the room. The glueball 1720 MeV = (11/6)·938 is a real forward prediction (Section 7) built on C_2^int (the labeled integer), not on the boundary gap 2(d_S−1) — so it cannot corroborate the gap value (GATE-KOIDE). Kept as a prediction; out of the room for the pin.

Tier: gap VALUE = DERIVED on the compact boundary — mass^2 = 2(d_S − 1) = 6, dimension-rooted, R-independent (K1713). The Δ→mass^2 shift is CLOSED (K1713 — the transverse gluon eigenvalue *is* the physical mass^2; no conformal shift). Three named residuals remain (stated up front, Section 0), none touching the value on the compact boundary: (1) the flat-$\mathbb{R}^4$ construction — that this compact-boundary gap is the $\mathbb{R}^4$ Yang–Mills gap (the Clay map exists); (2) the literally-YM identification — that the reconstructed edge theory is *interacting* SU(3) YM, not a gapped generalized-free cousin (Sections 6/9); (3) the constant-holonomy zero mode — the A_θ constant (m=0) mode is a finite-volume modulus, declared, not part of the gap. *(Residuals 1 and 2 are the two faces of the flat-space Clay residual; 3 is a finite-volume artifact.) Guard 1 holds: this is the gap of a boundary operator on D_IV^5; the $\mathbb{R}^4$ identification is residual (1), never fused.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — "you treat only SU(3)" — is answered by showing G is not a free input. Paper B establishes D_IV^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing dim_C = 5 and N_c = 3 (the dimension-free output of the proved T1829). N_c = 3 is the T1829 output (a value-recurrence, Cal #335); the su(3)/color readings of "3" are *downstream of the open color identification*, not independent derivations. So SU(3) is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix rank = 2 (N_c = rank^2−1), and rank generates {n_C, C_2, g} (T2491). The "all G" wall is dissolved.

4. W1 folded: the D_IV^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_IV^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_IV^5 carries the 4D conformal group SO(4,2) ⊂ SO(5,2) (the same edge where light and electromagnetism live):

OS datum	D_IV^5 realization	tier
rigorous Hilbert space	the Hardy space H^2(D_IV^5)	SOLID
Hamiltonian with a gap	form (Hodge) Laplacian on $(S^4\times S^1)/\mathbb{Z}_2$; gauge-form (1,0) pure-spatial; gap = mass^2 = 2(d_S−1) = 6, dimension-rooted, R-independent (K1713)	DERIVED (residuals: $\mathbb{R}^4$ construction, literally-YM, zero-mode)
reflection positivity	tube-type (Cayley → T(Ω)); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	SO(4,2) ⊂ SO(5,2), UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap Δ > 0 (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is SO(5,2)-equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* SO(5,2) positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. *(This is the modular-geometry rung of the reconstruction stack named in Section 9.)*

6. The one residual: interacting Yang–Mills, or generalized-free? (= “is the Rehren edge literally YM?”)

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting SU(3) Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3's naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056's naming (Rehren-edge = literally-YM): Rehren's algebraic holography is a rigorous bulk↔boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. "Genuinely interacting" and "literally the physical YM net" are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. "Non-commutative ==> interacting" is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n-point functions ($n \geq 3$). The YM self-interaction is the vertex [A,A], nonzero precisely because the gauge algebra is non-abelian. So

interacting <=> the gauge algebra is genuinely non-abelian su(3) <=> the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian su(3) (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^PC tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal's trim of the v0.2 over-claim "derived / landed interacting").

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D\_IV^5)$. Its bilinears F⊗F decompose into exactly four irreducible J^PC components, each of which must have eigenstates for the substrate's operator algebra to close:

channel	operator	what it commits
0^{++}	$Tr(F^2)$	energy density (the action)
0^{-+}	$Tr(F \tilde{F})$	topology (Pontryagin density)
2^{++}	Tr(F^$\mu\rho$ F^ν_ρ) traceless	curvature (the stress tensor)
1^{+-}	Tr(F[D,F])	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The oddballs (0^{+-}, 1^{-+}, 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the SO(2) dilatation) on the holomorphic discrete series on $H^2(D\_IV^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda\_0 + (\text{energy step})$, $\lambda\_0 = \text{genus}(D\_IV^5) = n\_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = SO(5) harmonic degree = spin J; the parity-odd (Hodge-dual) sector carries the half-canonical twist n_C/2. With one dimensionful anchor (seat = $\pi^5 \cdot m\_e = 156.4$ MeV; $m(0^{++}) = c\_2 \cdot \text{seat} = 1720$ MeV):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	J = 2	7/5	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	n_C/2	3/2	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	1 + n_C/2	17/10	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the 2^{++} = g/n_C leg is fully blind (genus + spin only, nothing read from the data, lands because g = n_C + rank). The 0^{-+} and 1^{+-} use the half-canonical twist n_C/2, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder λ_0 = n_C; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation 0^{++*} = (1,1) sits at the linear conformal energy $E = \lambda\_0$

+ 2 = 7, degenerate with 2^{++} (2408 MeV; lattice ~2670, within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel = 60 = C_2·n_C·rank); with the established χ_top^{1/4} = 180 MeV the WV identity gives f_G(0^{-+}) ≈ 12.6 MeV. Decay-side: mixing (glueball ↔ $q\bar{q}$ overlap on the shared Hardy space), not a "dump."

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg 2^{++} = g/n_C; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^PC tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade* / *proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The so(7) unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: su(3) ⊂ g_2 ⊂ so(7), and so(7) is the compact dual isometry of Q^5 — the same so(7) whose Casimir spectrum is the glueball tower of Section 2; g = 7 is the so(7) vector = g_2 fundamental = $3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain (su(3) not-subset-of so(5,2)) but is a geometric subalgebra of the compact-dual so(7); on the domain side it is the operator algebra on H^2. The discrete-series spectrum of this so(7) has the primaries {N_c, n_C, C_2, g} as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally SU(3) YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a mass gap whose derived compact-boundary value is mass^2 = 2(d_S−1) = 6 (dimension-rooted, R-independent, K1713).

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values → Bisognano–Wichmann-1975 / Borchers-2000 modular geometry → Rehren-2000 holographic net → Osterwalder–Schrader reconstruction → cluster decomposition.

Rehren's algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_IV^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical SU(3) YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded ⇒ gap) with no free dimensionless parameter beyond the ruler R, and its compact-boundary value is DERIVED: $\text{mass}^2 = 2(d\_S-1) = 6$, dimension-rooted, R-independent (Section 2, K1713 — the $\Delta\to\text{mass}^2$ shift is closed); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8). The glueball 1720 MeV at 0.6% is a real, separate forward prediction (built on C_2^int, not the gap $2(d\_S-1)$) but out of the room for the value-pin. What is materially advanced but still named-open: the cross-channel spectrum (Section 7). The flat-space residuals (1)–(2) are the construction + interacting/literally-YM identification (Sections 6/9); residual (3) is the constant-holonomy zero mode (a declared finite-volume modulus). This is a sharpening of the Clay problem to a derived compact-boundary value plus three named residuals — *not* a claim of its solution, never "YM proved."

Explanatory summary (Guards held). BST *derives the Yang–Mills mass-gap value on a bounded domain*: $\text{mass}^2 = 2(d\_S-1) = 6$, dimension-rooted (Guard 1: a boundary operator on D_IV^5, not the $\mathbb{R}^4$ Hamiltonian; Guard 3: dimension-rooted, not "colors," not the scalar Casimir), and the flat-$\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- $\Delta\to\text{mass}^2$ shift — CLOSED (K1713): no conformal shift; the transverse gluon eigenvalue *is* the physical mass^2. *(The Round-15/16 "three opens" are all resolved: reading the field as a form fixes the operator, keeps the pure-spatial (1,0) mode, and removes the R-dependence.)*
- Residual (3) — the constant-holonomy zero mode (declared): the A_θ constant (m=0) mode is a finite-volume modulus (the Polyakov holonomy), not part of the gap; declared as a finite-volume artifact, to be handled in the $\mathbb{R}^4$ / decompactification limit.
- The one flat-space residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) (algebra closure verified; identification in progress).

- Polyakov/Wilson-line mode at 10 vs deconfinement (Elie, stretch): the holonomy mode sits at 10, above the gap 2(d_S−1) = 6 — a candidate tie to the confirmed deconfinement T_c. Check the ordering and the T_c relation.
- D_A dynamical-gap check (Elie): does the fluctuated Dirac D_A^2 carry the same gauge-form gap dynamically (mass^2 = 2(d_S−1) on the real boundary; state the space first)? Report "the D_IV^5 gap"; do not relabel 2(d_S−1) as "C_2."
- Falsifier: the cross-channel glueball spectrum is a forward prediction — 0^{++*} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the derived gap value 2(d_S−1) = 6 and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v1.0-candidate (K1713 — the gap value CLOSED and dimension-rooted): reading the field as a 1-form fixes the operator as the form (Hodge) Laplacian, whose lowest transverse eigenvalue on the boundary sphere is 2(d_S−1) = 6, set by the sphere dimension d_S = n_C−1 = 4 — not colors (U(1)/SU(3)/SU(5) all give 6; the gauge group sets the degeneracy). The form parity keeps the pure-spatial (1,0) mode and gauge invariance eats A_θ, so mass^2 = 2(d_S−1) = 6, R-independent — a derived value on the compact boundary. Label discipline (Guard 3): 2(d_S−1), never "colors," never "= C_2" (the scalar Casimir; the morning's identity, K1707). The Δ→mass^2 shift is closed. The scalar readings (4, the (1,1) §570 candidate) were the wrong sector, superseded. Three named residuals: (1) the flat-$\mathbb{R}^4$ construction, (2) the literally-YM identification, (3) the constant-holonomy zero mode (declared finite-volume modulus). The Polyakov mode at 10 sits above the gap (deconfinement tie). The glueball 1720 stays separate (on C_2^int) — out of the room. The value is closed; the package is the deliverable. NOT "YM proved" (residuals (1)–(2) are the large $\mathbb{R}^4$ open), NOT "colors," NOT "= C_2." INTERNAL. Nothing pushed; CP existence-only.

Draft v1.0-candidate (K1713 — value closed, dimension-rooted). Spine: the field-is-a-form reading (gauge = 1-form → form Hodge Laplacian → 2(d_S−1) = 6, dimension-rooted, dimension-generic, R-independent via A_θ gauge-eaten); Grace's form-$\mathbb{Z}_2$ / dimension-genericity pin; Elie's transverse-mode close + Δ-shift-closed + the Polyakov-10 deconfinement tie + the zero-mode-as-modulus; Lyra's write-up + label discipline (dimension-rooted 2(d_S−1), not colors, $\neq C_2$); the so(7) unification; Paper B (gauge-group derivation); Cal (2(d_S−1) label, three residuals) + Keeper K1713 (color→dimension correction, Δ closed, zero mode named). The scalar readings (Rounds 15/16) and the "color-rooted 2N_c" (v0.9) were superseded. Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v1.0-candidate, K1713: gap value CLOSED — mass^2 = 2(d_S−1) = 6, dimension-rooted (boundary-sphere dimension d_S=4), R-independent, derived on the compact boundary (gauge field = 1-form → Hodge Laplacian → 2(d_S−1); A_θ gauge-eaten → pure-spatial (1,0)). Labeled 2(d_S−1), never "colors," never "= C_2". Δ-shift closed. Three residuals: flat-$\mathbb{R}^4$

construction, literally-YM, constant-holonomy zero mode. Glueball separate-valid, out of the room. The value is closed; the package is the deliverable.